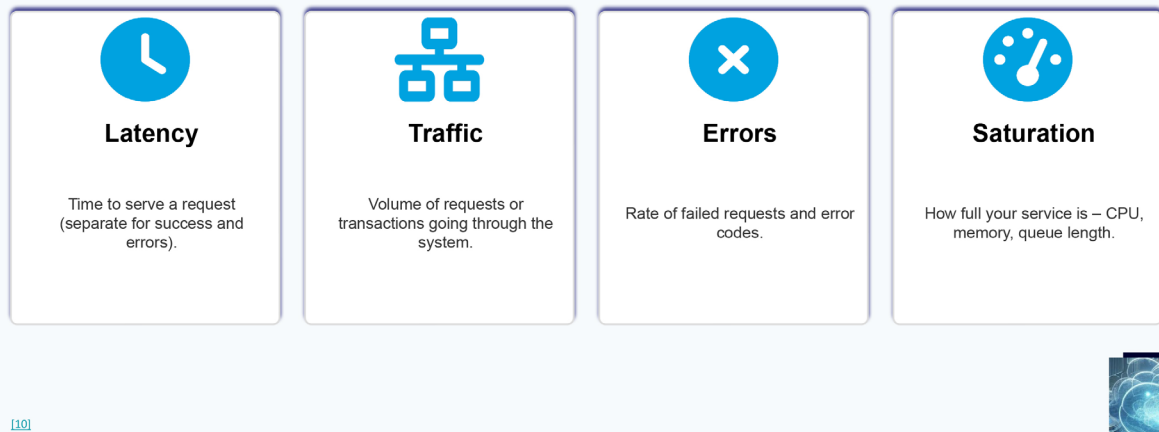


Golden Signals



The golden signals of observability

As modern applications are distributed across cloud platforms, Kubernetes clusters, databases and interconnected services, failures can occur across multiple components. The goal of chaos engineering is not to allow business-impacting failures but to proactively understand failure scenarios and strengthen system reliability. It has also become an important practice in Site Reliability Engineering (SRE), where concepts such as SLOs (Service Level Objectives), SLIs (Service Level Indicators) and SLAs (Service Level Agreements) help measure, maintain and improve system reliability.

The challenges Kubernetes faces

Kubernetes environments face several operational challenges, with scalability, performance and resource management being among the most critical. Since Kubernetes workloads are highly dynamic, organisations need effective autoscaling mechanisms such as Horizontal Pod Autoscaler (HPA) and Vertical Pod Autoscaler (VPA) to efficiently manage changing demands.

Debugging and root-cause analysis remain complex due to the distributed

nature of Kubernetes. Identifying issues across running pods, containers and clusters requires deeper visibility into system behaviour beyond basic monitoring.

Security and compliance are also major concerns, requiring additional tools and configurations to identify vulnerabilities and strengthen protection. Solutions such as Trivy, Falco and Snypk help improve security monitoring, while configuration management requires separate focus beyond observability.

Other key challenges include cost optimisation, service mesh complexity and visualisation. Optimising CPU and memory allocation helps prevent resource overprovisioning and reduces costs. Service mesh solutions such as Istio provide visibility into service-to-service communication, while visualisation tools help teams understand system behaviour through graphical interfaces rather than relying only on command-line tools.

The golden signals of observability

The golden signals of observability provide the foundation for monitoring modern applications and addressing Kubernetes challenges.

The first golden signal is *latency*, which measures the time taken to process requests and helps identify performance bottlenecks within the system. By analysing latency, organisations can understand delays in processing requests and optimise application and network performance.

The second signal is *traffic*, which provides visibility into the volume of requests entering the system and helps organisations understand workload patterns. For example, an e-commerce platform needs to differentiate between users who are browsing products and those performing critical activities such as placing orders or completing transactions, as each activity creates different demands on system resources.

The third signal is *errors*, which helps identify the type and frequency of failures occurring within the system. By analysing HTTP response codes such as 5xx, 4xx and 2xx, organisations can understand whether services are functioning as expected and identify recurring issues that require attention.

The fourth signal is *saturation*, which measures how efficiently the system handles incoming requests by analysing resource utilisation,